

Document Name: Master Batch Record: Downstream Purification (Affinity Capture and Anion Exchange)			Page 1	
Number: MBR-DSP-201	Version: 1.0	Effective Date: 01APR2026	Status: Effective	Part No: DS-OBX-201

Product and Lot Information

Product		Drug substance lot	
Process step		Manufacturing date	
Affinity resin		Affinity column ID	
AEX medium		AEX column ID	
Clarified harvest lot		Target percent full	

Referenced SOPs

SOP Number	Title
SOP-3300	Affinity Capture Chromatography (CaptureSelect AAVX)
SOP-3320	Low-pH Elution and Neutralization
SOP-3400	Anion Exchange Full and Empty Separation
SOP-3410	Fraction Collection and Pooling
SOP-1203	Buffer and Solution Preparation
SOP-1318	In-Process UV, Conductivity, and pH Monitoring
SOP-0905	Bioburden and Endotoxin Sampling

1. EQUIPMENT / INSTRUMENTS			
Description	EIN	Calibration Due	Performed By / Date
Chromatography skid			
Inline UV detector			
Inline conductivity monitor			
Inline pH meter			
Fraction collector			
Analytical balance			

2. BUFFER PREPARATION				
Buffer	Composition	Lot Number	Prepared By / Date	Verified By / Date

Affinity equilibration	20 mM sodium phosphate, 2 mM MgCl ₂ , pH 5.0			
Affinity elution	100 mM sodium citrate, 2 mM MgCl ₂ , pH 3.0			
Neutralization buffer	1 M Tris base, pH 8.5			
AEX equilibration	20 mM Bis-Tris-propane, pH 9.0			
AEX elution (high salt)	20 mM Bis-Tris-propane, 1 M TMAC, pH 9.0			

3. AFFINITY CAPTURE CHROMATOGRAPHY

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
3.1	Record clarified harvest load volume.	Load volume (L). Acceptance: record.			
3.2	Record equilibration conductivity.	Conductivity (mS/cm). Acceptance: 8 to 12.			
3.3	Confirm baseline stable before load.	Baseline stable?	☐ Yes ☐ No		
3.4	Record affinity elution pH.	Elution pH. Acceptance: 2.8 to 3.2.			
3.5	Record low-pH hold time before neutralization.	Hold time (min). Acceptance: not more than 15.			
3.6	Record neutralized pool pH.	pH. Acceptance: 7.0 to 8.0.			
3.7	Record affinity eluate pool volume.	Volume (L). Acceptance: record.			

4. ANION EXCHANGE FULL AND EMPTY SEPARATION

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
4.1	Record anion exchange load conductivity.	Conductivity (mS/cm). Acceptance: not			

		more than 5.0 to bind.			
4.2	Record anion exchange load pH.	pH. Acceptance: 8.8 to 9.2.			
4.3	Confirm gradient method loaded.	Method AEX-QA-01. Empty peak about 6.3 mS/cm; full peak about 10.1 mS/cm.	☐ Yes ☐ No		
4.4	Record fraction-collection (pooling) start conductivity.	Conductivity (mS/cm). Acceptance: not less than 8.5 to collect the full-capsid peak and exclude the empty shoulder.			
4.5	Record fraction-collection stop conductivity.	Conductivity (mS/cm). Acceptance: record.			
4.6	Record anion exchange pool volume.	Volume (L). Acceptance: record.			
4.7	Record anion exchange load total vector genomes.	Load VG (e13 vg). Acceptance: record.			
4.8	Record anion exchange pool total vector genomes.	Pool VG (e13 vg). Acceptance: record.			

5. POOL AND STEP RECOVERY					
Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
5.1	Calculate anion exchange step recovery.	Step recovery (percent). Acceptance: not less than 55 percent. Formula: Step recovery (percent) = (Pool VG / Load VG) x 100	Pool VG (e13 vg): Load VG (e13 vg): Result (percent):		
5.2	Record percent full capsids confirmed on the analytical	Percent full (percent), reference; confirmed on WS-			

	worksheet.	AEX-2605X. Acceptance: not less than 80.0.			
5.3	Confirm purified pool bioburden sample collected.	Bioburden sample collected?	☐ Yes ☐ No		

6. PROCESS COMMENTS			
Page / Step	Comment	Performed By / Date	Verified By / Date

BATCH RECORD APPROVALS	
Manufacturing Review	Quality Assurance Review

Manufacturing and QA review signatures are applied at batch disposition.